Task-Specific Action Editing for Household Manipulation: A Six-Task Development Pilot and a Twelve-Rollout Partial Submission

Author and affiliation pending confirmation

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Abstract

We document a paired development study of frozen released manipulation policies with task-specific action editing, together with a separate partial-submission evaluation. The development panel contains 36 physical rollouts: six tasks, three instances, and two policy conditions. Mean final goal progress falls from 0.455 to 0.329 with editing, and full success falls from 5/18 to 1/18. Task 001 improves partial progress without improving full success. The final panel fixes one edited policy per task for tasks 001, 002, and 005, with four public instances each. At this snapshot, 12/12 trials are complete; unfinished trials have no reported score. This report supports reproducibility and failure analysis rather than a claim of general policy improvement.

1 Scope and evidence

BEHAVIOR-1K provides realistic household activity simulation and goal-based evaluation [1]. The primary evidence here is the recorded paired pilot, not a new benchmark-wide result. Development instances are native 301 (public index 0) and training instances 23 and 24. These instances were used during development and must not be represented as an untouched held-out test. The supplementary final panel is a time-constrained partial submission. The user-approved execution order is 002, 001, 005; no further research rollouts are planned after its twelve trials.

2 Method and controls

We use a released task-specific policy implementation [4]. The original actor is frozen. Task-specific auxiliary networks were trained using 20 demonstrations, 1,000 updates, batch size eight, and discount 0.999. Each condition uses seed zero for the reported panel. The action-editing configuration uses a 30-step chunk, 15 executed steps per replan, carry length four, two original proposals and one edited proposal, one edit step, and edit scale 0.01. The approach is informed by action-editing and real-time policy work [2, 3]; these citations do not establish equivalence to their full algorithms or training claims. The paired panel reuses preserved historical baseline rollouts where recorded. It is not a prospectively randomized experiment. Capture alignment and environment/inference seeds are documented in the physical-row evidence. Different tasks use different released actors and auxiliary networks, so aggregate scores do not isolate a single universal editing effect.

3 Paired development results

Across 18 pairs, final Q averages 0.455 for the original actor and 0.329 for editing. Only task 001 shows a positive task-level mean Q difference (+0.222). Task 005 has approximately zero change, while task 007 decreases by 0.500. Full success decreases from five to one. Thus this configuration does not support an overall performance-improvement claim. Increasing partial progress can coexist with lower full success. Maximum intermediate Q is not interchangeable with final Q; ending predicates and retained goal satisfaction require separate video and metric inspection.

Task	A0 Q	RT Q	Delta Q	A0 success	RT success
000	0.667	0.333	-0.333	2/3	1/3
001	0.444	0.667	+0.222	1/3	0/3
002	0.238	0.143	-0.095	0/3	0/3
003	0.381	0.333	-0.048	1/3	0/3
005	0.444	0.444	+0.000	1/3	0/3
007	0.556	0.056	-0.500	0/3	0/3

Table 1: Three development instances per task and condition. Q is official final goal progress, not full task success.

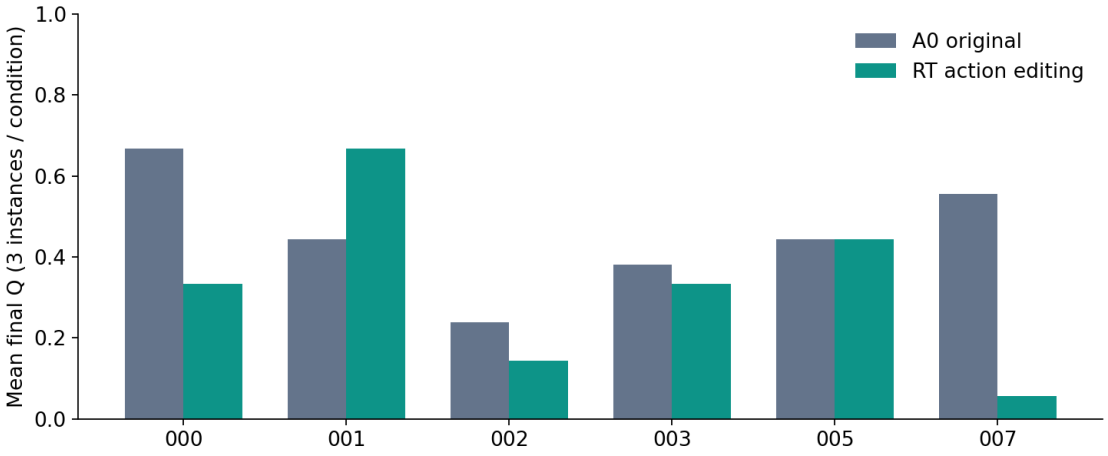


Figure 1: Mean final Q in the six-task paired pilot.

4 Final partial-submission protocol

The final panel contains public indices 0–3 once each for tasks 001, 002, and 005, totaling twelve trials. Native instance identifiers are 301–304. All started trials, including failures, must be retained. There is no selection of favorable repeats. Public-index-zero development exposure is disclosed; this report does not claim explicit organizer approval beyond the published protocol. The final policy remains frozen and has five supported routes (000, 001, 002, 003, 005); the evaluated subset is three routes. Preserving unevaluated routes avoids changing the executed policy identity. The execution manifest SHA256 is `f95003a2bdb8ce6f...` (full digest in the execution manifest).

Task	Public index	Snapshot state	Final Q
002	0	complete	0.571
002	1	complete	0.714
002	2	complete	0.571
002	3	complete	0.000
001	0	complete	0.000
001	1	complete	0.000
001	2	complete	0.000
001	3	complete	0.000
005	0	complete	0.000
005	1	complete	0.500
005	2	complete	0.000
005	3	complete	0.333

Table 2: Final-panel status at 2026-10-11T10:19:20 UTC. Pending and starting trials are unscored.

5 Runtime and reproducibility

The clean server has four NVIDIA L40S GPUs and 46 logical CPUs. Each GPU runs a policy and simulator. Simulator CPU quotas changed from seven to ten to 10.5 without restarting active task-002 trials; policy quota is four. These are shared cgroup ceilings, not exclusive core reservations. Per-trial quota-change records preserve this intervention. Timing comparisons must separate steady-state rollout throughput from startup, restoration, packaging, and backup. The pilot used a 224-pixel RGB DefaultWrapper. The final evaluator uses RGBDFullResWrapper and the unchanged R1Pro robot configuration. Input and runtime differences prevent direct attribution of pilot-to-final speed changes to hardware alone. The policy subsequently resizes RGB and does not use depth, while wrapper and simulator overhead still remain. The pinned evaluator revision is `bd049de3119acdcdf2334fe9e1ebe060fa20c108`. L40S memory use is not evidence that the release runs on the official single-24GB-GPU setup. The user explicitly elected to skip actual 24GB hardware validation, based on approximately 20GB observed GPU usage; no additional GPU rental is planned. Registry publication, organizer access, and portal submission remain pending. No additional research experiments are authorized after the twelve-trial panel.

6 Scoring and limitations

Official partial submissions count missing instances as zero. If q_i denotes each of the twelve final Q scores, the panel mean is $\sum_i q_i/12$, the selected three-task score with all ten public instances is $\sum_i q_i/30$, and the full 100-task denominator is $\sum_i q_i/1000$. These values should only be reported after all twelve trials complete, with denominators explicit. A mean over completed trials alone during execution is not the final panel score. The sample is small, tasks and final scope were selected during development, simulation is nondeterministic, and there is no matched final A0 control. No significance, generalization, or benchmark leadership claim follows from this evidence. Video analysis should distinguish navigation, grasp acquisition, manipulation, placement, and goal retention while linking each observation to its original artifacts.

7 Artifact organization and follow-up

The main repository presents the six-task pilot, final twelve-trial status, and submission instructions. The preparation branch preserves restoration, historical scope changes, and prior experimental procedures. Submission notes remain on main until the user explicitly requests their relocation after submission. GitHub branch privacy follows repository privacy. Private Hugging Face research and submission catalogs separate use cases. Large historical archives remain in their original private repository with pinned revisions and SHA256 receipts. Submission staging contains only completed unchanged raw metrics and videos; simulator data, decryption keys, and credentials are excluded. Metadata audits compare archive LFS SHA256 and download the small manifest, distinct from a new full archive redownload. The final portal requires organizer-accessible assets, not merely private owner-readable URLs. The next analysis uses preserved videos, metrics, inference logs, and goal traces rather than additional research rollouts. Author identity and affiliation require confirmation before publication.

References

[1] Chengshu Li et al.. BEHAVIOR-1K: A Human-Centered, Embodied AI Benchmark with 1,000 Everyday Activities and Realistic Simulation. 2024. `https://arxiv.org/abs/2403.09227`.

[2] Perry Dong, Kuo-Han Hung, Tian Gao, Dorsa Sadigh, and Chelsea Finn. EXPO-FT: Sample-Efficient Reinforcement Learning Finetuning for Vision-Language-Action Models. 2026. `https://arxiv.org/abs/2605.25477v2`.

[3] Perry Dong, Kuo-Han Hung, Dorsa Sadigh, and Chelsea Finn. Reinforcement Learning for Real-Time Vision-Language-Action Policies. 2026. `https://arxiv.org/abs/2609.18207v1`.

[4] Ilia Larchenko. BEHAVIOR-1K solution: released implementation. 2025. `https://github.com/IliaLarchenko/behavior-1k-solution/tree/ca556f74a455cef7987a2be4537b5ac85cc56dd7`.